INCH-POUND

MIL-H-44298C
30 June 1994
SUPERSEDING
MIL-H-44298B
3 September 1993

MILITARY SPECIFICATION

# HAM SLICES, CHUNKED AND FORMED, SMOKE FLAVORING ADDED, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers chunked and formed, smoke flavoring added ham slices, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

* 1.2 Classification. The product shall be of the following types as specified (see 6.1):

Type I - Ham Slices in Brine

Type II - Ham Slices in Spice Sauce

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5018, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8905/8940

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SPECIFICATIONS

MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-44340 - Can, Tray Pack

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks, and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations (9 CFR Parts 301-399)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

Institutional Meat Purchase Specifications for Fresh Pork, Series 400

(Copies are available from the Director, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2603, South Building, P.O. Box 96456, Washington, DC 20090-6456.)



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U.S. Standards for Condition of Food Containers

(Copies are available from the Chairperson, Condition of Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS) U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

* D 1974 - Methods of Closing, Sealing, and Reinforcing Fiberboard Shipping Containers
D 3330 - Peel Adhesion of Pressure-sensitive Tape
* D 5118 - Fabrication of Fiberboard Shipping Boxes

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)



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NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

* 3.2.1 Pork. Pork for use in preparation of canned, cooked and/or cured ham shall be derived from any one or any combination of the following USDA Certified Institutional Meat Purchase Specification (IMPS) Cuts: Item No. 402C (fresh pork ham, boneless, trimmed, weight range D), 402D (fresh pork ham, outside, weight range D), 402E (fresh pork ham, outside, trimmed, weight range C), and 402F (fresh pork ham, inside, weight range D). Tying of the fresh hams is not necessary. All pork shall be accompanied by a USDA, Agricultural Marketing Service (AMS) agent for condition of the product and compliance with the IMPS Quality Assurance Provisions.

NOTE: Certified IMPS Cuts may be further trimmed (see 3.3.1) and broken down into smaller cuts after certification, as requested by the purchaser. Boxes used for shipment of the further trimmed and broken down cuts shall be labeled as "Pork for further processing as specified in MIL-H-44298". The certificate shall be noted as such and shall accompany the shipment.

* 3.2.1.1 Handling and storage. Handling and storage of the boned and trimmed pork, prior to processing into the finished product, shall be in accordance with the following requirements:

a. Pork processed on the day of initial certification shall be maintained at an internal temperature of 28° to 50°F, inclusive.



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b. Holding in the fresh-chilled state for not more than 4 days after initial certification is permitted, provided that the pork is maintained at an internal temperature of 28° to 40°F, inclusive.

c. Holding in the frozen state for not more than 180 days after initial certification is permitted, provided that the pork is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence

d. Tempering/thawing of the frozen pork shall comply with the following:

- microwave tempering/thawing of pork is permitted. Any evidence of cooking, scorching, or browning caused by the microwave process shall be cause for rejection of he product which cannot be reworked or reoffered. Additionally, the product shall show no evidence of deterioration due to mishandling prior to thawing (for example, discoloration, foreign odor, excessive purge, or obvious deviation from specification handling and storage requirements) as determined by USDA
- the microwave tempering/thawing process shall not exceed 48 hours from time of removal of the pork from the freezer to further preparation
- regardless of the process used, frozen pork shall be tempered/thawed to an internal temperature not to exceed 40°F
- when a tempering/thawing process other than microwave is used, the time from the start of the tempering/thawing process removal from freezer) to further preparation shall not exceed 96 hours
- pork which has been tempered/thawed shall not be refrozen.

* 3.2.2 Cinnamon, ground. Ground cinnamon shall be prepared from the dried bark of Cinnamomum burannii Blume commonly known as "Korintiji cinnamon". The cinnamon shall contain not less than 1.5 mL of volatile oil per 100 grams of ground cinnamon and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

* 3.2.3 Clove, ground. Ground cloves shall be prepared from the dried, unopened flower buds of Caryophyllus aromaticus L. The powder shall be dark reddish-brown in color and shall possess a strong aromatic odor with a hot pungent taste. The ground cloves shall contain not less than 15.0 mL of volatile oil per 100 grams of ground clove and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.



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3.2.4 Flavoring, hickory smoke, single strength. Single strength hickory smoke flavoring shall be an oil-based liquid with a density of 7.6 pounds per gallon, a stability of greater than 30 hours (A.O.M.), and a phenolic level (dimethoxyphenol) of 7.3 mg. to 8.1 mg. per mL. The smoke flavoring shall have a mellow hickory smoke aroma (see 6.4.1).

* 3.2.5 Juice, lemon. Lemon juice may be canned or frozen, single strength or concentrated. The lemon juice shall be prepared from unfermented juice obtained from fresh, sound, mature lemons of one or more high acid varieties.

* 3.2.6 Margarine. Margarine shall conform to the FDA Standards of Identity for Margarine and shall possess a fine and pleasing flavor. The body shall be smooth, firm and homogeneous. The margarine shall possess a uniform medium yellow color and may be salted or unsalted.

* 3.2.7 Nutmeg, ground. Ground nutmeg shall be derived from dried kernel of Myristica fragrans Houtt. The nutmeg shall have a dull brown color. The ground nutmeg shall contain not less than 7.5 mL of volatile oil per 100 grams of ground nutmeg and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 20 sieve.

* 3.2.8 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this specification. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.2.9 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.9.1 Salt tablets. Salt tablets shall be noniodized, white, refined sodium chloride and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.10 Sodium ascorbate. Sodium ascorbate shall comply with the Food Chemicals Codex.

3.2.11 Sodium erythorbate. Sodium erythorbate shall comply with the Food Chemicals Codex.

3.2.12 Sodium nitrite. Sodium nitrite shall comply with the Food Chemicals Codex.

3.2.13 Sodium tripolyphosphate. Sodium tripolyphosphate shall comply with the Food Chemicals Codex.



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* 3.2.14 Starch, waxy maize, modified. Starch shall be white, odorless, finely pulverized, modified waxy maize starch for use in thermostabilized foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140°F to 170°F and be fully hydrated at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress and conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste, and shall comply with MIL-STD-900.

3.2.15 Sugar, white, granulated. Sugar shall be white, refined, granulated cane or beet sugar, or a combination thereof, and shall comply with MIL-STD-900.

3.2.16 Water. Water used for formulation, blanching, ice making, cooking, cooling, and washing shall conform to the National Primary Drinking Water Regulations.

3.3 Preparation and further processing. Processing shall be on a continuous basis.

3.3.1 Preparation of the cured ham. The pork shall be prepared as follows:

a. Boned and trimmed pork which meets the requirements of 3.2.1 shall be further trimmed, if necessary, to ensure compliance with the finished product requirements (see 3.6).

b. The boned and trimmed pork shall be reduced in size to chunk-type pieces by mechanically grinding once through a kidney-shaped plate, having openings measuring 2.5 inches or more in the least dimensions, and using a two-bladed knife that yields pieces weighing at least 2 ounces.

3.3.1.1 Mixing. The chunked pork pieces shall be mechanically mixed with the following cure ingredients for approximately 15 minutes or until the product becomes moderately tacky.

| Ingredient | Pounds per 100 pounds of ham |
|---|---|
| Water or ice | 3.0000 |
| Salt 1/ | 2.5000 |
| Sugar, white, granulated | 1.0000 |
| Sodium tripolyphosphate | 0.2500 |
| Sodium ascorbate and/or sodium erythorbate | 0.0550 |
| Flavoring, smoke, single strength | 0.0400 |
| Sodium nitrite 2/ | 0.0156 |

1/ The total amount of salt in the formula shall be adjusted, as necessary, to produce a product that complies with the finished product salt requirements (see 3.6).



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2/ Sodium nitrite shall be of sufficient strength to ensure a complete cure without exceeding 156 parts per million of nitrite in the cured ham during preparation.

The mixture shall be held prior to or after the stuffing operation at an internal temperature of 38°F to 42°F for not less than 4 hours nor more than 72 hours before cooking to ensure uniform cure dispersion in the mixture.

3.3.1.2 Stuffing and forming. The mixed meat shall immediately be tightly stuffed into casings of a type and size to accommodate the cooking method and the finished product meat dimension requirements. Alternatively, the mixed meat may be formed in stainless steel molds to a size which will ensure compliance with finished product meat dimension requirements.

3.3.1.3 Cooking. The formed ham logs shall be cooked by one of the following methods to provide a product meeting the finished product requirements:

a. Cookhouse method. The ham logs shall be cooked in a cookhouse (smokehouse without smoke) at a temperature range of 160°F to 162°F and 100 percent relative humidity until the internal product temperature reaches 150°F to 152°F.

b. Water cooking method. Water cooking is permitted only for the ham logs in casings provided they are cooked in water until the internal temperature of the log reaches 150°F to 152°F.

3.3.1.4 Cooling and storage. Immediately after completion of the cooking process, the cooked ham logs shall be placed in an ice water bath or cooled by other commercially acceptable cooling methods and rapidly cooled to 130°F or below. The ham logs shall be further cooled from 130°F to 80°F or below within two hours, and further cooled from 80°F to 40°F or below within four hours. The cooled, cooked ham logs shall be held for not more than 48 hours at an internal temperature of 28°F to 40°F after removal from the cooking process and prior to slicing. Alternatively, if the cooked ham logs are to be held for more than 48 hours, they shall be mechanically stuffed and vacuum-sealed in heat-sealable casing materials made from FDA and USDA food approved laminate consisting of 0.0005-inch-thick polyester and 0.0015-inch-thick medium density polyethylene. The cooled, cooked ham logs shall be maintained at an internal temperature of 28°F to 40°F for a period of not more than 60 days (canned or cooked pullman-style hams may be held for not more than 90 days) from the date of pack until slicing.

3.3.1.5 Slicing. The ham logs shall be sliced to a uniform size, shape, and thickness to ensure compliance with finished product requirements (see 3.6). The ham slices shall be filled into the tray pack cans within four hours after being sliced. End slices shall not be used.



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3.3.1.6 Alternate processing for cured ham. Cured hams processed by methods other than the chunking method described in 3.3.1(b), such as sectioned and formed, or massaging of intact ham muscle, shall be acceptable. If using an alternate processing method, the preparation and processing techniques of this specification, except for chunking, shall be required.

3.3.1.7 Alternate cured ham product Canned or cooked pullman-style hams which meet the requirements of 3.3.1 and 3.3.1.1 and can be sliced to the requirements of 3.3.1.5 shall be acceptable.

3.3.2 Brine preparation (for type I only). The brine shall be prepared in a manner to ensure compliance with finished product salt requirements. (At the time of filling, salt tablets may be substituted for granular salt in brine preparation.) The brine shall be made fresh daily, and the brine remaining at the end of a production day shall be discarded.

* 3.3.3 Sauce preparation (for type II only). The spice sauce shall be formulated and prepared as follows:

| Ingredient | Percent by weight |
|---|---|
| Water | 58.95 |
| Sugar, white, granulated | 28.00 |
| Margarine | 6.00 |
| Starch, waxy maize, modified | 5.10 |
| Juice, lemon (single strength) 1/ | 1.50 |
| Cinnamon, ground | 0.25 |
| Clove, ground | 0.15 |
| Nutmeg, ground | 0.05 |

1/ The total amount of lemon juice in the sauce formula shall be adjusted with water when concentrated lemon juice is used.

NOTE: The following sauce preparation procedures were used in the development of this product. Alternate procedures may be used provided finished product requirements are met. (When alternate procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable).

a. The starch and part of the water shall be mixed to form a slurry.

b. The balance of the water and the margarine shall be heated in a steam-jacketed kettle until the margarine is fully melted.

c. The balance of the ingredients, except for the starch slurry, shall be added and uniformly blended with the water and melted margarine.



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d. The starch slurry shall be added and the sauce shall be heated with constant stirring until the sauce temperature reaches 180°F to 190°F. The mixture shall be held for approximately five minutes in this temperature range.

e. The volume of the sauce shall be adjusted with water to compensate for evaporation loss during heating and holding.

f. The sauce shall be maintained in the temperature range of 150°F to 180°F and filled into tray pack cans within four hours after preparation.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product to conform to the finished product requirements and to the following requirements:

a. Not less than 54 ham slices shall be shingled in two rows, lengthwise, in the tray pack can.

b. The product at the time of filling shall be in the temperature range of 28°F to 40°F.

* c. For type I product, each tray pack can shall be filled with the product and brine to a total can content weight of 108 ounces. Brine may be filled hot or cold.

* d. For type II product, each tray pack can shall be filled with product and spice sauce to a net weight of 108 ounces.

e. Immediately after filling, each can shall be sealed in accordance with the can manufacturer's guidelines/requirements and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable (see 4.5.5), and under a vacuum established by a processing authority and specified in the scheduled process so as to ensure compliance with the finished product requirement (see 3.6y).

f. The filled and sealed tray pack cans shall be in the retort process within two hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value ($F_o$) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.



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b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

* d. No individual can shall have a total can content weight of less than 106 ounces for type I product.

* e. No individual can shall have a net weight of less than 106 ounces for type II product.

* f. The average total can content weight shall be not less than 108 ounces for type I product.

* g. The average net weight shall be not less than 108 ounces for type II product.

h. Drained weight of 54 intact ham slices in any individual tray pack can shall be not less than 56.0 ounces.

i. The average drained weight of the slices shall be not less than 58.0 ounces.

j. No individual can shall contain less than 54 intact slices.

k. Ham slices shall approximate the size and shape shown in figure 1 or 2.

l. The texture of the slices shall not be dry, rubbery, or mushy.

* m. For type I product, the average fat content of the drained product shall be not greater than 3.5 percent, and no individual tray pack drained product shall have a fat content greater than 5.5 percent.

* n. For type I product, the salt content of any individual tray pack drained product shall be not less than 1.5 percent nor greater than 2.5 percent.

* o. For type II product, the average fat content of the finished product shall be not greater than 5.5 percent, and no individual tray pack finished product shall have a fat content greater than 7.5 percent.

* p. For type II product, the salt content of any individual tray pack finished product shall be not less than 1.0 percent nor greater than 2.0 percent.

q. No individual can shall contain a bone piece measuring 0.3 inch or more in any dimension.



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r. No individual can shall have slices with adherent fat or a fat pocket measuring more than 0.5 inch by 0.5 inch.

s. No individual can shall have slices with a void area or air pocket measuring more than 0.5 inch by 0.5 inch.

t. Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual can shall be not greater than 2.0 ounces.

* u. No individual can shall have slices that contain finely ground, flaked, or otherwise comminuted product.

* v. For type II product only, the sauce shall not be lumpy.

* w. For type II product only, the viscosity of the sauce shall be not less than 7 cm per 10 seconds nor greater than 20 cm per 10 seconds when determined by a Bostwick Consistometer (see 4.5.3.4).

x. The product shall show no evidence of excessive heating (materially darkened or scorched).

y. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The pork component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Product Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.



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4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the AOAC International, and in the approved methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation, or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

* 4.5.3.1 Total can content or net weight examination. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in



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preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. For type I or type II product, if the average total can content or net weight is less than 108 ounces or if the total can content or net weight of any individual can is less than 106 ounces, the lot shall be rejected.

4.5.3.2 Double sampling plan for product examination. The finished product shall be examined for the defects listed in table I utilizing the double sampling plans indicated in MIL-STD-105. The lot size shall be expressed in tray pack cans. The sample unit shall be one filled and sealed tray pack can. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 6.5 for minor defects. The sample cans shall be heated in accordance with heating instructions on the can label.

TABLE I. Product defects 1/ 2/

| Category | | Defect |
|---|---|---|
| Major | Minor | |
| 101 | | Less than 54 intact slices in any individual can |
| 102 | | Drained weight of 54 intact slices in any individual tray pack can is less than 56.0 ounces 3/ 4/ 5/ |
| 103 | | Texture of the slice is dry, rubbery, or mushy |
| 104 | | Product shows evidence of excessive heating (materially darkened or scorched) |
| 105 | | Presence of a bone piece measuring 0.3 inch or more in any dimension |
| 106 | | Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual can is more than 2.0 ounces |
| 107 | | Slices not approximating the size and shape shown in figures 1 or 2 6/ |
| 108 | | More than five slices have adherent fat or a fat pocket measuring more than 0.5 inch by 0.5 inch |
| 109 | | More than five slices have a void area or air pocket measuring more than 0.5 inch by 0.5 inch |
| *110 | | More than five slices contain finely ground, flaked, or otherwise comminuted product |



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TABLE I. Product defects 1/ 2/ (cont'd)

| Category | | Defect |
|---|---|---|
| Major | Minor | |
| | 201 | Five or less slices have adherent fat or a fat pocket measuring more than 0.5 inch by 0.5 inch |
| | 202 | Five or less slices have a void area or air pocket measuring more than 0.5 inch by 0.5 inch |
| * | 203 | Five or less slices contain finely ground, flaked, or otherwise comminuted product |
| | 204 | Slices not arranged in a shingled manner |
| * | 205 | Sauce is lumpy (type II only) |

1/ The presence of foreign material (for example, glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot.

3/ For type I product, to determine drained weight, the contents of the tray pack can shall be poured into a U.S. Standard 1/2 inch sieve in a manner that will distribute the product over the sieve without breaking the ham slices. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a $45^{O}$ angle and allowed to drain for two minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

*4/ For type II product, to determine drained weight, the sauce in each can shall be poured off, strained through a U.S. Standard No. 8 sieve, and individually reserved for viscosity determinations. The remaining contents shall be poured into a flat-bottom container. A minimum of three times the tray pack can's volume of $180^{O}$F to $190^{O}$F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquefy rendered fat and remove the sauce without breaking the ham slices. The contents shall then be poured into the U.S. Standard 1/2 inch sieve in a manner that will distribute the product over the sieve without breaking the ham slices. Sieve area



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shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45° angle and allowed to drain for two minutes before determining the drained weight. Fifty-four ham slices shall be weighed to determine the drained weight. The drained weight shall be reported to the nearest 0.1 ounce.

5/ If the sample average drained weight is less than 58.0 ounces, the lot shall be rejected.

6/ Ham slices approximating the shape of figure 2 which have broken corners shall not be considered as intact slices.

* 4.5.3.3 Fat and salt content testing. Three filled and sealed tray pack cans shall be selected at random from the lot. The product shall be tested for fat content in accordance with the Official Methods of Analysis of the AOAC, methods 976.21, 985.15, or 960.39, and for salt in accordance with the Official Methods of Analysis of the AOAC, method 935.47, except that preparation of the samples shall be as follows: The unopened tray pack cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. For type I product, the tray pack cans shall be opened and the brine solution drained. The remaining drained contents of each can shall be separately blended in a Waring Blendor or equivalent. For type II product, the cans shall be opened and the entire contents of each can shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat or salt requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

* 4.5.3.4 Viscosity testing (for type II only). The strained sauce collected from each of the cans selected for drained weight inspection (see 4.5.3.2 and footnote 4/ to table I) shall be individually tested for viscosity as specified below (see 6.5).

Instrument: Bostwick Consistometer
Catalog number:
23270-004 or 15-347-50
VWR Scientific Company
P.O. Box 7900
San Francisco, CA 94120

Fisher Scientific
585 Alpha Drive
Pittsburgh, PA 15238

Method:

a. Level the instrument.

b. Bring sauce to 100°F ± 1°F in a water bath in a covered container.

c. Stir sauce thoroughly before filling the Bostwick cavity.



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d. Scrape sauce evenly across upper edge of cavity.

e. Release sauce and time sauce flow to the nearest one second and measure distance traveled.

If the Bostwick viscosity value of the sauce from any can in the sample fails to conform to the requirement specified in 3.6, it shall be classified as a major defect and the lot shall be rejected.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the U.S. Standards for Condition of Food Containers, except that the inspection for labeling shall be in accordance with 4.5.4.1. In addition, scratches, scuffs, or abrasions that occur on the outside coating as a result of the filling, sealing, and thermoprocessing of the tray pack cans shall not be scored as a defect.

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4) except, for self-adhering labels, the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings

Minor: Air bubbles under label
Label not properly adhered to can, for example, label raised or peeled back from edges or corners

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's guidelines/ requirements and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns shall be classified as a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to 75°F ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. When examining a ribbed lid, lay the straight edge only between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance shall be classified as a major defect and shall be cause for rejection of the lot.



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* 4.5.7 Shipping container examination. Thirteen shipping containers shall be examined for the defects listed below. The finding of any major defect or more than three minor defects shall be cause for rejection of the lot.

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

* 5.1.1 Level A. For type I or for type II product, one hundred and eight ounces of product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5.

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

* 5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with a HSC full depth cover, grade V2s of ASTM D 5118. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with ASTM D 1974. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1), except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.



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5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

* 5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of ASTM D 5118. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of class domestic, grade 175 fiberboard.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (for example, EST 38) or a three letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1994 would be coded as 4078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code. Following the four digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 end of the can. The labeling shall be legible when



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examined as specified in 4.5.4 after preparation of product in accordance with heating instructions. Paper labels are not permitted. In addition, cans shall show the following statements:

TO HEAT IN WATER: Submerge unopened can in boiling water. Simmer gently 25 to 30 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

TO HEAT IN OVEN: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven 30 to 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 3 slices each.

As an alternate labeling method, a preprinted, self-adhering 0.002 inch thick clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering labels are used, the tray pack can shall be labeled with the Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary markings in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE, UNLESS PALLET RACKS ARE USED.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.



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* b. Type of product required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.4, and 6.2).
e. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
f. Level of packing required (see 5.2).
g. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Ingredient information.

6.4.1 Flavoring, hickory smoke, single strength. It has been found that CharOil Hickory (single strength) produced by Red Arrow Products Company, P.O. Box 1537, Manitowoc, WI 54221-1537 meets the requirements of 3.2.4 and performs satisfactorily in this product. The 0.0400 pounds per 100 pounds is based on the phenolic levels specified in 3.2.4. The cured ham producer is cautioned to verify these requirements prior to formulation of the ham cure ingredients.

* 6.5 Alternate viscosity testing method. The contracting officer may authorize an alternative contractor-recommended method of viscosity testing if the alternative method is approved by the U.S. Army Natick Research, Development and Engineering Center.

6.6 Subject term (key word) listing.

Canned food
Combat field feeding
Operational ration
Food processing
Shelf stable



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* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 35

Preparing activity:

Army - GL

(Project 8905-1131)

Review activities:

Army - MD, QM
Navy - MC
DLA - SS



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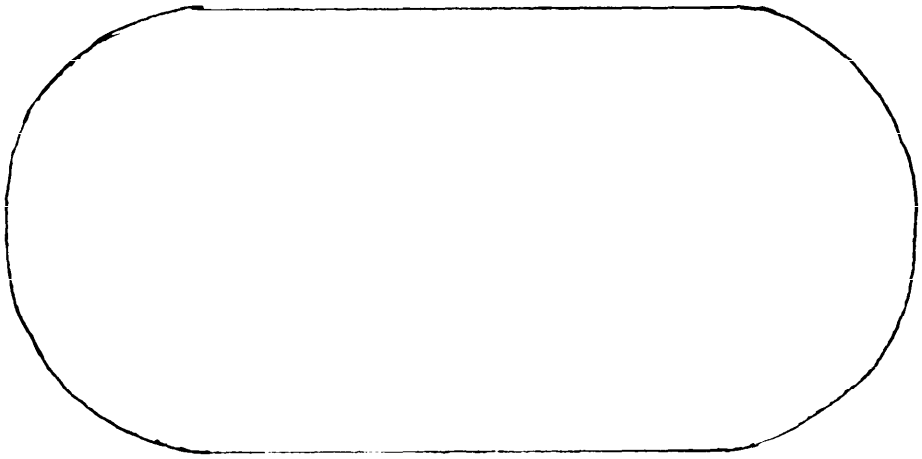


Figure 1. Ham Slice Shape

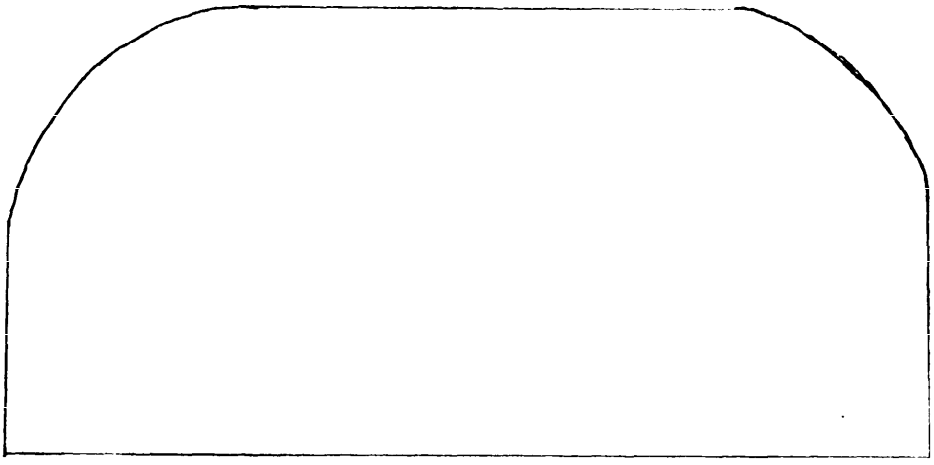


Figure 2. Alternate Ham Slice Shape



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-H-44298C | 94-06-30 |

3. DOCUMENT TITLE
Ham Slices, Chunked and Formed, Smoke Flavoring Added, Thermostabilized, Tray Pack

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial | (2) AUTOVON/DSN |
|---|---|---|
| U.S. Army Natick RD&E Center | 508-651-4501 | 256-4501 |
| c. ADDRESS (Include Zip Code)<br>Commander, U.S. Army Natick RD&E Center<br>ATTN: SATNC-WRE<br>Natick, MA 01760-5018 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

DD Form 1426, OCT 89 NATICK OP4, JUL 91 *Previous editions are obsolete.* 198/290

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